



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

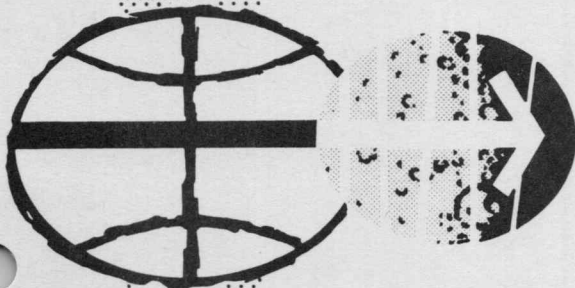
APOLLO 17

DECEMBER 7 LAUNCH

THIS ISSUE IS FOR THE DECEMBER
7 LAUNCH. DO NOT DISCARD
PREVIOUS ISSUES.

CHANGE AA (T+24)
CSM CUE CARDS

PREPARED BY
FLIGHT PLANNING BRANCH
CREW PROCEDURES DIVISION



MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

NOVEMBER 22, 1972

APOLLO 16
CSM CUE CARDS

Change A

NOVEMBER 22, 1972 (T+24)

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CSM CUE CARDS

DECEMBER 7 LAUNCH

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 CHANGE B 11/15/72
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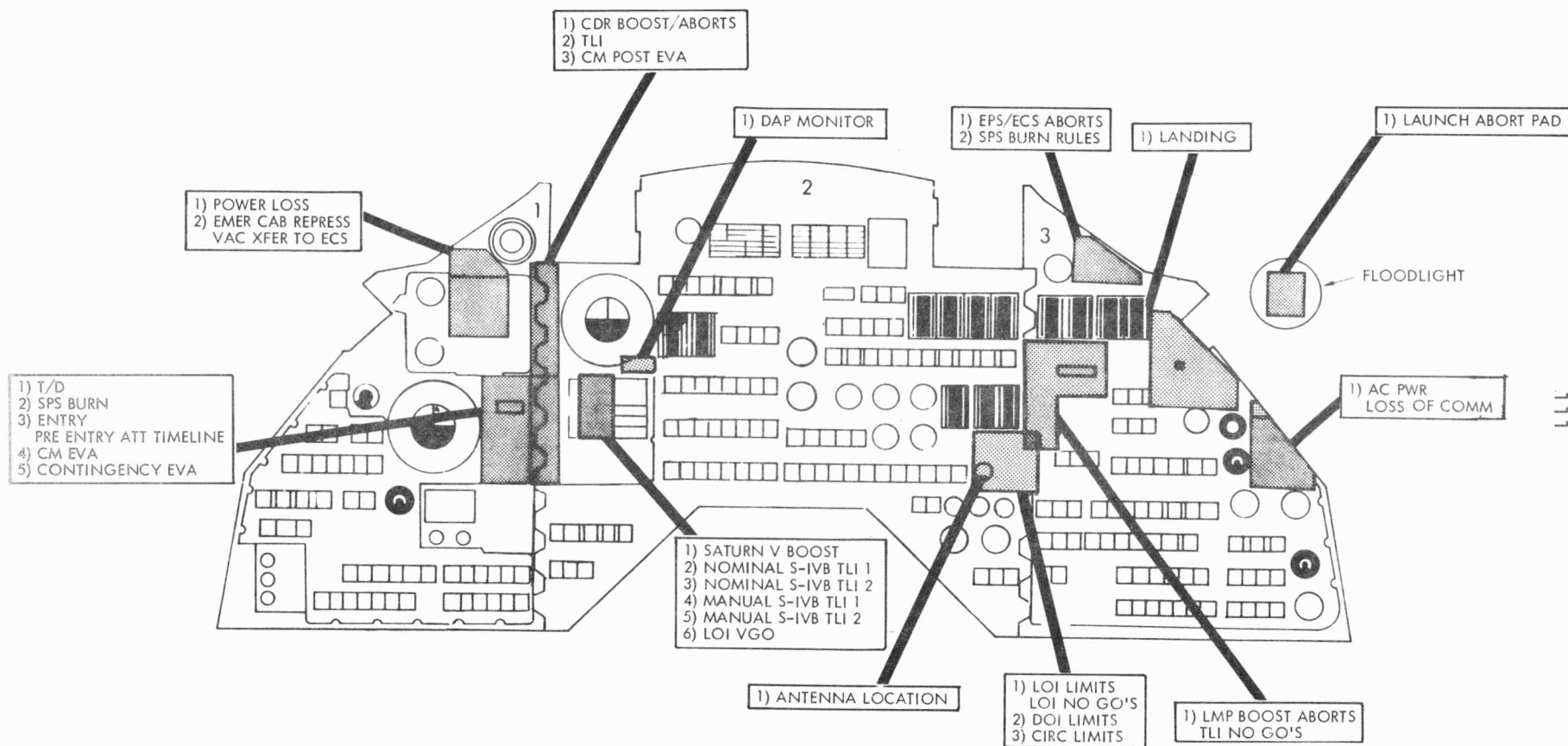
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*Current Changes, data

LIST OF EFFECTIVE PAGES (CONT)

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	VELCRO LOCATIONS (BACK OF CARD)	(Dec 6, T=0)
*3-4	LOI LIMITS	11/22/72 (T+24)
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3-22	PNL 229 cb FUNCTIONS	(Dec 6, T=0)
3-23	PNL 229 cb FUNCTIONS CONT'D	(Dec 6, T=0)
*3-24	LOI VGO	11/22/72 (T+24)
	VELCRO LOCATIONS (BACK OF CARD)	(Dec 6, T=0)
3-25	DOI 1 LIMITS	(Dec 6, T=0)
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3-27	CIRC LIMITS	(Dec 6, T=0)
3-28	VELCRO LOCATIONS (BACK OF CARD)	(Dec 6, T=0)

*Current Changes, data



CSM CUE CARD LOCATIONS

9/4/72

L1A

L1B

CDR BOOST/ABORTS

DEC 6/7 LAUNCH 9/4/72

EMS - ΔV TO 6999.9 FPS
 -3:00 DSKY - VERIFY P02
 V75 DO NOT ENTER

00:00 LIFT OFF, MODE 1A
 CLOCK RUNNING
 P11 AUTO START
 00:02 YAW MNVR
 00:12 R&P PROG
 00:30 ROLL COMPL
 00:50 MONIT q_α TO
 2:00, 100%, 5°ATT ERR
 CABIN PRESS DECR
 01:01 MODE IB
 PRPLNT DUMP - RCS CMD
 01:23 MAX Q
 01:57 MODE IC
 02:00 NO COMM - FEET WET
 EDS AUTO - OFF
 2 ENG OUT - OFF
 LV RATES - OFF
 V82 N62
 02:19 IECD
 02:41 OECD/STAGE/IGN
 03:19 TWR JETT, MODE II
 α /Pc-Pc
 MAN ATT (P)-RT CMD
 STEAM/H2O - AUTO
 03:22 GUIDE INIT (OECD +41)
 ELS/SEP cb's open
 05:55 SIVB TO COI
 06:00 GMBL MTRS - ON
 CK GPI
 P-0.58°
 Y+1.90°
 07:41 IECD
 08:11 PU SHIFT
 08:36 LEVEL SENSE ARM
 09:17 MODE IV
 09:20 OECD/STAGE/IGN
 10:24 MODE III
 11:51 SECD
 12:01 INSERT VI

THC C/O VI+100

+4°/sPY
 +20°/sR

+9°/sPY
 +20°/sR

MODE IA (:00 - 1:01)

:00 CCW NEUT *CM/SM SEP*

:14 ELS AUTO

ELS LOGIC

TWR-APEX JETT

:16 DROGUES

:18 He DUMP

<3.8K OR 28s & <10K - MAINS

MODE IB (1:01 - 1:57)

:00 CCW NEUT *CM/SM SEP*

:11 CANARDS

:14 ELS AUTO

ELS LOGIC, RCS CMD-ON

24K TWR-APEX JETT, DROGUES

MODE IC (1:57 - TWR JETT)

:00 CCW NEUT *CM/SM SEP*

RCS CMD-ON

:11 CANARDS

Ha>32&PLAT GO

TWR JETT

MAN ATT(P)-

RT CMD

0/135/0

BMAG(3)-1/2

EMS-ENT/NORM

.05G LT/SW-ON

MAX LIFT

<32 or NO GO

*P+5°/SEC *

HI+P, R 90°

*DAMP RATES *

ROLL HDS DN

30K ELS LOGIC

ELS AUTO

24K TWR-APEX

DROGUES

MODES II - III - IV

:00 CCW (4s)

:03 *CSM/LV SEP - PUSH*

*RCS CMD-ON *

:05 NEUT, +X TO :24

II N44(TFF)

TFF>2m: Y 45L

BMAG(3)-1/2

BAT C

ELS/SEP close

SM SEP

CM RCS

0/120/0

LM FNL SEP

EMS-ENT/NORM

.05G LT/SW-ON

MAX LIFT

III N50(ΔR)

BMAG(3)-1/2

180/194/0

EMS-NORM

BURN ΔR=0

TFF>2m: Y 45L

BAT C

ELS/SEP close

SM SEP

CM RCS

0/105/0(FL)

LM FNL SEP

EMS-ENT/NORM

.05G LT/SW-ON

AT .2G

305/105/0(HL)

IV N62(VI)

180/356/0

EMS-NORM/IGN

H>70 +6s or

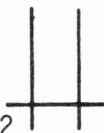
H=200 & +HDOT

9/4/72



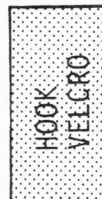
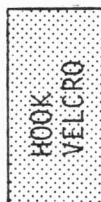
L1B

10/13/72



L1A

10/13/72



L2

9/4/72

L2

SATURN BOOST

DEC 6/7 LAUNCH

11/2/72

DET	Θ	VI	$\dot{H}$	H
00:00	90	1341	0	.0
:30	85	1397	286	.6
1	68	1867	786	3.2
1:30	50	3001	1454	8.7
2	35	5000	2160	17.6
2:19	29	6804	2688	25.2
2:30	26	7782	2894	30.2
2:41	23	8966	3130	35.8
3	23	9133	2758	44.8
3:30	24	9664	2274	57.2
4	22	10295	1853	67.4
4:30	20	11024	1464	75.5
5	18	11850	1109	81.9
5:30	15	12776	793	86.5
6	13	13811	518	89.8
6:30	10	14965	290	91.7
7	7	16257	117	92.7
7:30	4	17708	10	93.0
8	6	19127	-29	92.9
8:30	1	20434	-10	92.8
9	358	21795	38	92.9
9:20	355	22777	112	93.1
9:30	355	22861	67	93.3
10	350	23395	-5	93.4
10:30	347	23957	-50	93.3
11	344	24546	-64	93.0
11:30	341	25163	-44	92.7
11:51	341	25599	-6	92.6

HOOK
VELCRO

SHIM - CONTINUOUS STRIP OF 4 PLY

HOOK
VELCROHOOK
VELCRO

CSM CUE CARDS

1-3

LOCATION
MDC-2 covering a
portion of DSKY lights

L3

LMP BOOST/ABORTS

DEC 6/7 LAUNCH 9/4/72

00:00 IA
 00:50 CAB PRESS DECR
 01:01 IB
 01:57 IC
 02:41 OECO/STAGE/IGN
 03:19 TWR JETT, II
 STM/H20 - AUTO
 06:00 ✓GMBL MTRS ON
 06:15 OMNI D
 (C if AZ>96)
 09:17 IV
 09:20 OECO/STAGE/IGN
 10:24 III
 11:51 SECO
 12:01 INSERTION

IA <3.8K; MAINS

IB 24K: TWR JETT
Ck LDG PHASE

IC HA>32 & PLAT GO
 TWR JETT
 MAN ATT P - RT CMD
 0/135/0
 BMAGS 1/2
 EMS - ENT/NORM
 .05G .05G on
 Max lift

<32 or NO GO
 * P rate +5°/s *

Ck LDG PHASE

II BMAGS 1/2
 cb MNA&B BAT C
 cb ELS/CM-SM SEP
 SM SEP
 0/120/0
 cb LM FNL SEP
 LM SEP
 EMS - ENT/NORM
 .05G .05G on, EMS ROLL
 Max lift
 Ck LDG PHASE

III BMAGS 1/2
 ΔR>0: 180/194/0
 (BEF/UP/HORIZ)
 ✓SCS TVC P&Y - AUTO
 EMS - NORM
 ΔV THRUST A - NORM
 2:05 DIR ULL & THRUST ON
 Burn ΔR = 0
 ΔV THRUST (2) - OFF
 cb MNA & B BAT C

SM SEP
 0/105/0
 (BEF/DN/Full lift)
 cb ELS/CM - SM SEP
 cb LM FNL SEP
 LM SEP
 EMS - ENT/NORM
 .05G .05G on, EMS ROLL
 .2G 305/105/0, Half lift
 Ck LDG PHASE

CUT OUT
THIS PART

IV BMAGS 1/2
 180/356/0
 FIXED ATT:
 (SEF/DN/HORIZ)
 ✓SCS TVC P&Y - AUTO
 EMS - NORM
 ΔV THRUST A - NORM
 FIXED ATT:
 (IGN @ +2:05)

<1:30 DIR ULL & THRUST ON
 Fly HDOT = 0
 HP >70 +6s BT
 or HA = 200 & +HDOT
 TFF<1+40: OFF
 HP Stop: OFF
 ΔV THRUST (2) - OFF

CUT OUT
THIS PART

TLI NO GO'S

11/2/72

LOSS OF:

ECS: CABIN

FIRE or
SMOKE

O2 MANF (Leak)

(2) MAIN REG

PRIM COOLANT LOOP

HOOK
VELCRO

CRY: O2 or H2 TANK

EPS: (2) FUEL CELL

ENTRY BAT

MN BUS

BAT BUS

AC BUS

BAT RELAY BUS

(2) INV

PRIM RAD

GLYCOL

(Leak)

HUMIDITY

HIGH

(2) SUIT

COMP

SUIT CIR

(3) QB DUMP

HOOK
VELCRO ϕ A AC1 or 2

SEC: SMJC (Activated)

SEQ SYSTEM

G&N: AUTO ATT (1) Axis

RATE DAMP (1) Axis

DIR RCS (1) Axis

(2) BMAG R, P or Y

(2) FDAI

THC

G&N: (2) RHC
CMCHOOK
VELCRO

ISS

(2) TVC SERVO LOOP

(2) DSKY

SPS: FU or OX TANK (Leak)

(2) GN2 TANK (Leak)

(2) BALL VLV BANK

(2) FEEDLINE T <40F

FU/OX Δ P >20

He TANK (Leak)

HOOK
VELCRO

RCS: SM He TANK (Leak)

Leak below ISOL VLV

(2) PKG T <55 F

(2) P or Y THRUSTER

(3) R THRUSTER

(2) CM He TANK (Leak)

& BLOWDOWN

(ARMED <1250)

CM MANF (Leak)

HOOK
VELCRO

L4

HOOK
VELCROHOOK
VELCRO

9/4/72

MODES

H I V AK

L4

LAUNCH ABORT PAD

9/4/72

✓✓✓*	TIG SECO + 2:05	
	EMS ΔV	6 9 9 9 9
*✓✓✓	ΔV FOR HP >70	—
	CUTOFF	
✓✓✓	BT	
✓✓✓	P (IGN)	
✓✓	GET 300K	
✓✓	P (.05G)	
✓✓	GET DRO	

*TIG AT APOGEE

*ΔV FOR 3350 NM SPLASH

L5

EPS/ECS ABORTS

9/4/72

EPS ABORTS

UNCONTROLLED MAIN BUS SHORT
LOSS OF ALL SM PWR SOURCES & 1 BAT
LOSS OF BOTH AC BUSES DURING I & II

CUT OUT
THIS PART

LOSS 2 OR 3 FC

ATTEMPT RECONNECT
EDS AUTO - OFF
USE BAT C & AUX

BAT BUS SHORT >22a

EDS AUTO - OFF
cb BAT BUS/MN BUS - OPEN
cb BAT C/MN BUS - CLOSE

MN BUS SHORT >25a

EDS AUTO - OFF
USE BAT C, INV 3
PWR DN SHORTED BUS

IF SEP REQ: AUTO RCS

TVC GMBL DRIVE - 2
SPS GMBL MTR P2 & Y2 - START
SPS GMBL CNTL cb (4) - OPEN

LOSS 1 AC BUS OR INV

USE INV 3

LOSS BAT RELAY BUS

NO ACTION

LOSS 1 OR 2 BATS

EDS AUTO - OFF
USE BAT C
cb BAT BUS/MN BUS - OPEN

ECS ABORTS

LOSS OF CABIN & SUIT PRESS
CABIN PRESS & SUIT 02
NOT SUFFICIENT

L5

10/13/72

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

LOCATION
MDC-3 adj to
VHF antenna sel

AC PWR

9/4/72

AC BUS
LIGHT

+

MN BUS UNDER-
VOLT

+

BUS OVERLOAD

CUT OUT
THIS PARTAFFECTED AC BUS - OFF
WITHIN 5s (AC BUS SHORT)

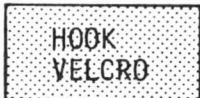
MN BUS LOSS

BRING ON INV 3
RECONFIGURE FC'sHOOK
VELCRO**AC USERS**SUIT COMP, GLY PUMP, TELECOM,
FC PUMPS, G&N PWR, S-BD XPND/PWR AMP,
RAD FLOW CONT**ELECTRICAL POWER USERS**02 HTRS, GMBL MTRS, RAD HTRS, SM RCS HTRS,
POT H2O HTR, H2 HTRS, CABIN FANS, SPS LINE HTR,
LIGHTS, TAPE RCDR, GLY/EVAP STM PRESS,
H2O FLOW CONT, TEMP IN CONT, CMC TO STBY, G&N PWR DOWN,
SEC LOOP, ECS RAD CONT/HTR cb's, SCE PWR, VHF, HGA PWR,
ONE INVERTER OPERATION, INSTRU ESS MNA & B cb's,
CRYO FANS, SPS GAUGING SWITCH, TVC GIMBAL DR (P&Y) -1,
BMAG 2-OFF, C/W NORMAL - ACKLOCATIONMDC-3 on right side
of DC AMP & VOLT ind

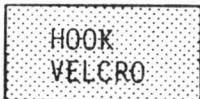
LOSS OF COMM

9/4/72

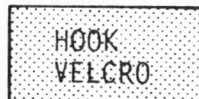
- A. Verify comm
Awake Configuration
- B. UP TLM CMD - RESET
Verify All cbs Pnl 225
closed

HOOK
VELCRO

- C. Attempt reacquisition with HGA if signal strength <60%.
- D. Select omni desired if unable to acquire with HGA.
- E. Verify AUDIO CTR & Personal COMM EQUIP Operation
- F. If still no comm, XPNDR & PWR AMP - SEC
- G. Wait 90 sec. If still no comm, PMP PWR-AUX.
- H. If still no comm, S-BD AUX TAPE-DN VOICE BU
- I. Request ground transmit CREW ALERT
- J. Go to T/C MALF.

HOOK
VELCRO

When CREW
ALERT ON,
UP TLM -
UP VOICE BU

HOOK
VELCRO

(Wait 3 min for MSFN reconfig.)

L7

10/13/72

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

PIA

PIB

TLI

11/22/72

11/22/72

DEC 7 LAUNCH

SIVB SEP

1. MAN ATT (3) - RATE CMD
THC PWR - ON
RHC PWR - NORM/DIRECT
RCS CMD - ON
S/C CONT - SCS
LV XLUNAR - SAFE
2. cb SECS ARM (2) - CLOSE
SECS LOGIC (2) - ON
SECS PYRO (2) - ARM
3. THC +X & HOLD (5 sec)
CSM/LV SEP - PUSH
4. OR - THC CCW (HOLD 5 sec)

O>F:ΔP>36
F>O:ΔP>26
OX:P>50

TLI

EMS ΔV TEST, SET ΔV
cb DIRECT ULL (2) - CLOSE
UP TL - BLK
EDS PWR - ON
PYRO ARM (2) - ON
THC, RHC
SC CONT - SCS
LV IND - SIVB
LV GUID - IU
XLUNAR - INJECT
SET DET : 51:00

*LV GUID LT - ON
*LV GUID - CMC
*RHC DIR (2) - OFF
*V16N20
*MNVR to R2 ALIGN

TB6 (TIG -9:38)
SII SEP LT, UPLK - ON
-9:00 SII SEP LT OUT
START DET UP
(TB6 1 REV IF SAFE)
51:04 (SAFE DELAY TIL 58:36)
51:05 REPRESS START
54:00 ORDEAL 300/LUNAR
56:00 ORDEAL $\theta = 17^\circ$
*ORDEAL $\theta = 0$ (N20)
*R2 ALIGN = 272°
 (57:00) ORDEAL - OPR
 $\theta = 13^\circ$
*R2 IGN 265.5°
58:15 AVG G (MNVR)
58:20 TVC SERVO 1 - AC1/MNA
EMS - NORMAL
58:36 SII SEP LT - ON
(TB6 1 REV IF SAFE)
58:38 SIVB ULLAGE
59:42 SII SEP LT - OFF
(INHIBIT: STG TIL :12)
59:52 FUEL LEAD
59:55 ORDEAL $\theta = 6^\circ$
*ORDEAL $\theta = 0$
*R2 IGN = 265.5°
ULLAGE OFF
59:59 LV ENG 1 LT - ON
*P = 0, Y = 00°

00:00 TIG
VI BT
00:02 LV ENG 1 LT - OFF
00:05 ORB RT OFF - IGN ON
01:32 PU SHIFT
05:46 SIVB ECO, TB7
05:56 LV ENG 1 LT - OFF +10
08:26 SIVB MNVR ORB RT (.3)

MONITOR

10°/s: BMAG 1 - STG C/O
45° P or Y: STG C/O
20°/s R
SIVB GUID LT: CONT - CMC,
GUID CMC, V46 STG C/O
NO ECO @ TFC = 0 + 1 sec or
 $\Delta tB + 1$ sec or EMS -150:
THC - CCW, NEUT in 1 sec

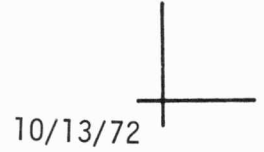


P1B

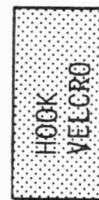
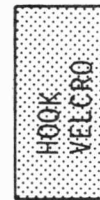
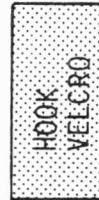
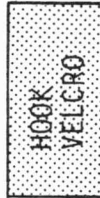


10/13/72

P1A



10/13/72



P2X

P2X

11/15/72

NOMINAL SIVB TLI 0

DEC 7 LAUNCH(T+24)

11/22/72

DET	Θ	Ψ	VI	H	H
0:00	272	359.7	25610	-3	95
:30	266	359.3	26161	-13	95
1	264	359.0	26768	8	95
1:30	263	358.5	27402	82	95
2	262	357.9	28160	216	96
2:30	261	357.4	28965	429	98
3	260	356.8	29813	727	100
3:30	259	356.3	30709	1121	105
4	258	355.7	31658	1617	112
4:30	256	355.2	32668	2225	121
5	254	354.7	33748	2948	134
5:30	250	354.2	34917	3774	150
5:47	250	354.2	35642	4307	162

PRINT FLIGHT CONFIGURATION
CARDS ON K10



11/15/72

MANUAL SIVB TLI 0

DEC 7 LAUNCH(T+24)

11/22/72

DET	Θ	Ψ	VI	H	H
0:00	266.0	356.5	25610	-3	95
:30	264.9	356.5	26161	-13	95
1	263.8	356.5	26768	8	95
1:30	262.8	356.5	27402	82	95
2	261.7	356.5	28160	216	96
2:30	260.6	356.5	28965	429	98
3	259.5	356.5	29813	727	100
3:30	258.4	356.5	30709	1121	105
4	257.4	356.5	31658	1617	112
4:30	256.3	356.5	32668	2225	121
5	255.2	356.5	33748	2948	134
5:30	254.1	356.5	34917	3774	150
5:47	253.6	356.5	35642	4307	162

LOCATION

MDC-2 covering a
portion of DSKY lights



P3X

P3X

CSM CUE CARDS

P2

11/22/72

NOMINAL SIVB TLI 1

DEC 7 LAUNCH

11/22/72

DET	Θ	Ψ	VI	$\dot{H}$	H
0:00	272	359.9	25605	-4	97
:30	265	358.7	26156	-15	97
1	264	359.1	26765	5	97
1:30	262	359.3	27401	77	97
2	261	359.4	28159	209	98
2:30	260	359.5	28964	418	99
3	259	359.6	29813	713	102
3:30	258	359.7	30710	1102	106
4	257	359.8	31661	1593	113
4:30	255	359.9	32674	2196	122
5	253	0.0	33759	2914	135
5:30	250	0.1	34933	3734	151
5:47	249	0.1	35648	4255	163

11/22/72

MANUAL SIVB TLI 1

DEC 7 LAUNCH

11/22/72

DET	Θ	Ψ	VI	$\dot{H}$	H
0:00	265.5	0	25605	-4	97
:30	264.4	0	26156	-15	97
1	263.3	0	26765	5	97
1:30	262.2	0	27401	77	97
2	261.1	0	28159	209	98
2:30	260.0	0	28964	418	99
3	258.9	0	29813	713	102
3:30	257.8	0	30710	1102	106
4	256.7	0	31661	1593	113
4:30	255.6	0	32674	2196	122
5	254.6	0	33759	2914	135
5:30	253.5	0	34933	3734	151
5:47	252.2	0	35648	4255	163

P3

P3

LOCATION

MDC-2 covering a
portion of DSKY lights

CSM CUE CARDS

2-4

P4

P4

11/22/72

NOMINAL SIVB TLI 2

DEC 7 LAUNCH

11/22/72

DET	Θ	Ψ	VI	$\dot{H}$	H
0:00	270	359.9	25600	-5	98
:30	264	358.0	26225	-15	98
1	262	359.5	26931	15	98
1:30	261	0.8	27674	110	99
2	260	2.0	28455	278	100
2:30	259	3.1	29276	529	102
3	258	4.3	30141	869	105
3:30	257	5.4	31055	1308	110
4	256	6.4	32026	1855	118
4:30	254	7.5	33060	2514	129
5	252	8.5	34167	3290	143
5:30	249	9.1	35365	4160	161
5:36	249	9.1	35629	4361	166

11/22/72

MANUAL SIVB TLI 2

DEC 7 LAUNCH

11/22/72

DET	Θ	Ψ	VI	$\dot{H}$	H
0:00	264.2	4.5	25600	-5	98
:30	263.2	4.5	26225	-15	98
1	262.1	4.5	26931	15	98
1:30	261.0	4.5	27674	110	99
2	259.9	4.5	28455	278	100
2:30	258.8	4.5	29276	529	102
3	257.8	4.5	30141	869	105
3:30	256.7	4.5	31055	1308	110
4	255.6	4.5	32026	1855	118
4:30	254.5	4.5	33060	2514	129
5	253.4	4.5	34167	3290	143
5:30	252.4	4.5	35365	4160	161
5:36	252.2	4.5	35629	4361	166

LOCATION
MDC-2 covering a
portion of DSKY lights

HOOK SHIM - CONTINUOUS STRIP OF 4 PLY VELCRO

HOOK SHIM - CONTINUOUS STRIP OF 4 PLY VELCRO

P5

P5

CSM CUE CARDS

T&D

9/4/72

SIVB Mnvr to SEP ATT: TB7 +15 min

_____:_____:_____: SIVB MNVR

_____:_____:_____: SEP

_____/_____/_____: RPY SEP ATT

_____/_____/_____: RPY EXTR ATT

DIR 02, CAB PRESS ~5.7

cb DOCK PROBE (2) - close

CUT OUT
THIS PART

COAS PWR - on

RCS DAP 11103/01111

V46E

N17 (SEP) & N22 (EXTR)

V63E (SIVB mnvr)

PROBE EXTD/REL - RETRACT

SM RCS tb(8) - gray

AUTO RCS SEL (16) - MNA/MNB

EMS NULL BIAS CK, -100/ΔV

FDAI - 5/1

MAN ATT (3) - RATE CMD

LIMIT CYCLE - OFF

ATT DB - MIN, RATE - LOW

THC PWR - on (up)

RHC NORM (2) - AC/DC

RHC DIR (2) - MNA/MNB

ATT SET 0/180/0

Set up TV

CMC MODE - FREE

SC CONT - CMC

BMAG (3) - RATE 2

cb RCS LOGIC(2) - open

TVC SERVO PWR #1 - AC1/MNA

Set DET 59:30

FC REAC vlv - LATCH

V49E F 06 22 (EXTR)

THC & RHC - ARMED

cb SECS LOGIC (2) - close

cb SECS ARM (2) - close

SECS LOGIC (2) - on (up)

OMNI C

RCS CMD - ON

HBR/RCD/FWD/CMD RESET

SECS PYRO ARM (2) - ARM

GDC ALIGN, EMS MODE - NORM

V62E

59:30 Start DET

59:50 CMC MODE - AUTO

59:58 Thrust +X & hold

00:00 CSM/LV SEP pb

00:03 THC release ~0.5 fps

Ck SM RCS PRPLNT, He &
FUEL PRESS

Ck 02 TK 3 ISOL vlv

MAN ATT(P) - ACCEL CMD

00:15 Pitch up ~0.5°/sec

When P error needle +:

PRO F 50 18 OMNI B

PRO 06 18

MAN ATT(P) - RATE CMD

F 50 18 Mnvr Compl

ENTR

THRUST +X ~4 sec, ~.7 fps

cb DIR ULL (2) - open

RCS DAP 11102/01111

Start TV & DAC

Stab/align CSM

BMAG (3) - 1/2

CAPTURE PROBE tb - bp

*Remains gray: *

*PROBE RETR-PRIM 1 *

*(SEC 1)in that sys *

CMC MODE - FREE

Damp ~10 sec

Align P&Y with THC <3°

PROBE RETR - PRIM 1 (2 alt)

SECS PYRO ARM (2) - SAFE

SECS LOGIC (2) - OFF

EDS PWR - OFF

cb EDS (3) - open

PROBE EXTD/REL - OFF

PROBE RETR (2) - OFF

cb DOCK PROBE - open

TAPE RCDR - off (ctr)

PCM BIT RATE - LOW

DAC & TV - off (ctr)

LOCATION

MDC-T covering ABORT
and LV light panel

P6

10/13/72

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

SPS BURN

10/13/72

EMS ΔV CK, Set ΔV , EMS FUNC - ΔV
Ck SIM BAY

GMBL TRIM

BMAG (3) - RATE 2

P _____

AUTO RCS SELECT - as req

Y _____

LOAD DAP

Set DET, SC CONT - CMC/AUTO

MNVR to PAD BURN ATT

BORESIGHT & SXT STAR CHECK, V41 N91E

P40 to F 50 18, GDC ALIGN

HOOK
VELCROCUT OUT
THIS PART

cb. STAB CONT (Pn1 8) & SPS(12) - close

MAN ATT (3) - RATE CMD

ATT DB - MIN, RATE - LOW

SCS TVC (2) - RATE CMD

 ΔVCG - LM/CSM or CSM

TVC GMBL DR P & Y - AUTO

54:00 *MN BUS TIE (2) - on.

(-06:00) *TAPE RCDR - HBR/RCD/FWD/CMD RESET

SPS He v1vs (2) - AUTO $\checkmark$, Ck N2A & N2B

*TVC SERVO PWR 1 - AC1/MNA, 2 - AC2/MNB

RHC PWR NORM (2) AC, DIRECT (2) - OFF

*BMAG (3) - 1/2

SC CONT - SCS, RHC 2 - ARMED

55:00 PRIMARY TVC CHECK

(-05:00) *GMBL MOT P1 & Y1 - START

Verify trim control & set, Verify MTVC

SCS: SCS TVC (2) - AUTO

SC CONT - CMC(SCS), THC - CW, Verify NO MTVC

SEC TVC CHECK

*GMBL MOT P2 & Y2 - START

Set GPI trim, Verify MTVCTHC neutral, Verify NO MTVC

Verify GPI returns to 0,0 (CMC) or trim (SCS)

RHC PWR NORM(2)-AC/DC

DIRECT (2) - MNA/MNB

(TRIM) BMAG (3) - RATE 2

PRO

BMAG (3) - 1/2

ENTR

HOOK
VELCROHOOK
VELCROHOOK
VELCRO

LOCATION

MDC-1 covering ABORT
and LV light panel

10/13/72

SPS BURN CONTD

*F 50 25 GMBL TEST OPTION

(ACCEPT) PRO

06 40 FDAI SCALE - 5/1

RATE - HIGH (IF SCS: After IGN), Update DET

59:00 *EMS - NORM, THC PWR - ON

(-01:00) *ΔV THRUST A(B) - NORM, THC & RHC (2) - ARMED

59:30 *Ck PIPA BIAS <2 fps for 5 sec

(-00:30)

59:XX *ULLAGE _____

(-00:XX)

HOOK
VELCRO

F 99 40 (-00:05) ENG ON ENABLE REQUEST

00:00 IGN, If SCS: THRUST PB - PUSH

00:03 BT >10 sec: ΔV THRUST B(A) - NORMAL

00:XX ECO

F 16 40 ΔV THRUST (2) - OFF

GMBL MOT (4) - OFF

TVC SERVO PWR 1 & 2 - OFF

F 16 85 Null residuals

RHC & THC - LOCKED

THC PWR - OFF, RHC PWR DIRECT (2) - OFF

cb DIRECT ULLAGE (2) - open, cb SPS P1 & Y1 - open

Record ΔV counter & residuals

EMS FUNC - OFF, MODE - STBY

PRO (MINKEY: To Seq 3X.2), ATT DB - MAX

BMAGS (3) - RATE 2, MN BUS TIE (2) - OFF

PCM BIT RATE - LOW, Ck SIM BAY

HOOK
VELCRO

THRUST FAIL ROUTINE

EXIT 9740 E 0640 9940 E 1685

RECYCLE 9740 E 0640 9940 P 0640

ERR NEEDLES 9740 P 0640

HOOK
VELCRO

G&N Reignition

F 97 40

ENTR

F 99 40

ΔV THRUST

✓GMBL TRIM

ULLAGE

PRO for IGN

HOOK
VELCRO

SCS Reignition

Check ATT

BMAGS

TVC (2) - AUTO

SC CONT - SCS

EMS - ΔV/NORM

ΔV THRUST

ULLAGE & THRUST

K2

SPS BURN RULES

9/4/72

FU/OX TEMP >40 (45-75°F)
FU/OX PRESS >115* (170-195 psi)
FU/OX ΔP <20 psi
Pc >70* (95-105 psi)
N2 A&B >400 (2900 psi max)

*TIGHT:
>160 & >80

CUT OUT
THIS PART

BURN: Start watch, INJ VLVS (4) - OPEN
He VLV tb (2) - gray, PUGS - balanced

SPS PRESS 1t

Continue critical burn

FU/OX LO: He VLV (2) - ON, FU/OX HI: He VLV (2) - OFF
HI ΔP: He VLV(2) - ON, If persists - OFF till Pc <70*

EARLY SHUTDOWN

He VLV (2) - ON
cb PILOT VLV (2) - close ✓
cb He VLV (2) - close ✓
cb EPS GP 3 & 5 - close ✓
ΔV THRUST (2) - NORM

NO SHUTDOWN

ΔV THRUST (2) - OFF
THC - CW
THRUST DIR ON - OFF
cb PILOT VLV (2) - open
cb EPS GP 5 - open

K2

10/13/72

HOOK
VELCRO

HOOK
VELCRO

HOOK
VELCRO

LOCATION
MDC-3 on left side
of VHF ANTENNA Sel sw

K3

LOI LIMITS

DEC 7 LAUNCH

11/22/72

VG0	BT	VM	MODE	ABORT LIMITS	CALL OUT MODES AND LIMITS
2980	00	0	I DPS @ 2hr	TIGHT	EARLY CUTOFF
2180	1+58	800		TK <160 & LO Pc ΔP >20 & LO Pc Pc <80 or DECR >10 & LO ΔV	OUT OF LIMITS: NO RESTART LIMITS GOOD: RESTART
2175	2+03	835	I DPS @ 30m		
			I DPS @ 30m +APS @ 2.5hr NO COMM: APS ASAP	LOOSE	EARLY CUTOFF
1975	2+31	1035		TK <115 & LO Pc Pc <70 & other cues	OUT OF LIMITS: NO RESTART LIMITS GOOD RESTART
1580	3+21	1400	II DPS @ 2hr +DPS @ PC		
			III DPS @ PC	BALL VLV FAIL, SHUTDOWN GOOD BANK 10 sec PRIOR to NOM C/O	
1262	4+03	1718			
0	ECO 6+35	2980	III	TIGHT	

RESIDUALS: DO NOT TRIM
MAX IGN DELAY: 2+00

C/O: BT+10 sec, IGN DELAY ED:
1+00 : BT+5, 2+00 : BT

LOI NO GO'S

11/2/72

LOSS OF:**ECS: CABIN**

FIRE or SMOKE

02 MANF (Leak)

(2) MAIN REG

PRIM COOLANT

PRIM RAD

GLYCOL (Leak)

HUMIDITY HIGH

(2) SUIT COMP & VAC CLNR

SUIT CIR

(3) OB DUMPS

CRY: 02 or H2 TANK**EPS: (2) FUEL CELL**

ENTRY BAT

MAIN BUS

BAT BUS

AC BUS

BAT RELAY BUS

(2) INV

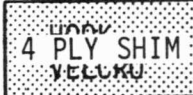
 ϕ A AC1 or 2

(4) DOCK LATCH

SEQ: SMJC ACTVD or SEQ SYS**G&N: AUTO ATT (1) Axis**

RATE DAMP (1) Axis

DIR RCS (1) Axis


 4 PLY SHIM
VELCRO


 HOOK
VELCRO
G&N: (2) BMAG R, P or Y

(2) FDAI

(2) RHC

CMC & DPS

ISS

O-DAC & DPS

(2) TVC SERVO LOOP

or (1) & DPS

(2) DSKY & DPS

SPS: FU or OX TANK (Leak)

(2) GN2 TANK (Leak)

or (1) & DPS

(2) BALL VLV BANK

or (1) & DPS

FEEDLINE T <40F

FU/OX Δ P >20

Pc <70

FLG T >480F

He TANK (Leak)

RCS: (2) SM He TANK (Leak)

(2) Leak below ISO VLV

(2) PKG T <55F

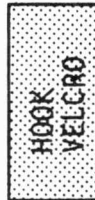
(2) P or Y THRUSTER

(2) R THRUSTER

(2) CM He TANK (Leak)

& BLOWDOWN (ARMED <1250)

CM MANF (Leak)


 HOOK
VELCRO

ENTRY

11/2/72

43:00(17:00) P62
 HRZN CK:(31.7±5)P _____+5°
 SCS, YAW LEFT (315°),
 BMAGS - ATT 1/RATE 2

45:00(15:00) SEP (Y=315°)
 MIN IMP/RATE 2, YAW TO 0
 FDAI - 5/5 PITCH TO .05G ATT (152)

R _____ P _____ Y _____

.05G FILTER
 - down ✓

CUT OUT
 THIS PART

_____ : _____ MOONSET GET _____ : _____

59:00(-1:00) PITCH NEEDLE OFF PEG, CK
 DET

00:00 RRT _____ : _____ : _____

_____ : _____ .05G

+3 NO .05G LT & P64 RUNNG
 EMS - BU/VHF RNG

+5 NO .05G LT & NO P64
 CK CLOCK

+5 .05G LT ON & NO P64
 FLY EMS

CMC GO/RT CMD/SC CONT-CMC

NO DAP/RT CMD/FLY BETA

NO CMC/RT CMD/FLY EMS

1.5G LIFTUP

G MAX SYS OFF >1G-NO GO

5.5G NO ROLL RVS SCS-EMS

_____ : _____ V CIRC

*DL _____ *VL _____

_____ : _____ DROG GET _____ : _____

P63

06 64 .0G. VI., RTG.0

V16 63 RTG.0, VIO.,
 TFE (MON)

KREL -5:28(55:00)

59:00 PITCH NEEDLE OFF PEG

06 64 RTGO → EMS

.05G CMC RET.05G±5
 ↓ (EMS .05G+3:MAN)

P64 .05:(L DN: CMC CMD,
 PAD, NO COMM)

06 74 BE.TA, VI., .0G
 1.5 L UP IF SHALLOW

MAX _____ ±1G. METER
 (EMS)

>6.5 >90° ROLL (EMS)

DO _____ MODULATE

P65 IF 26 < VL < 18

16 69 BE.TA

.DL _____ / _____ (EMS)

PRO VL. _____ / _____ (EMS)

06 74 BE.TA, VI., .0G

V16 64 .0G, VI., RTG.0(EMS)

N68 BETA, VI, HDOT

P66 <.2G KREL

06 22 RPY HRZN 31.7±12°
 <.05G ATT HOLD L UP,
 ✓ N22

EXIT P66

V16 68 BE.TA, VI., HDOT.
 NEG (IF +:EMS)

P67 .2G

06 66 BE.TA, XER.R, DNER.R

V16 64 .0G, VI., RTG.0(MON)

3400:STEAM PRESS

1900:N67

1700:50K

16 67 RTG.0, L.AT, LO. NG

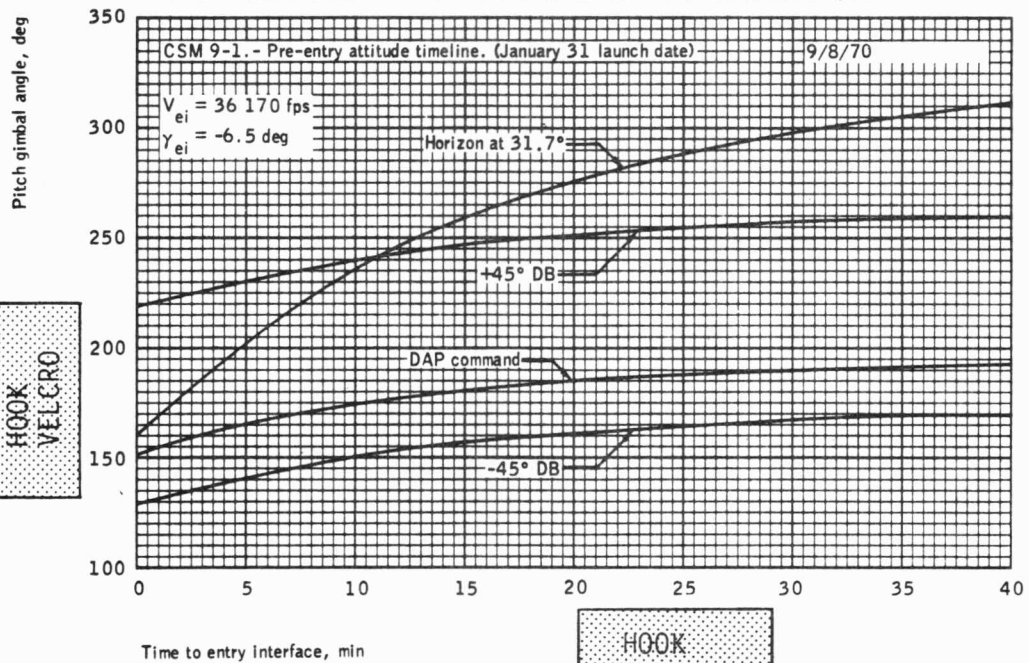
LOCATION

MDC-1 covering ABORT
 and LV light panel

10/13/72

HOOK
VELCROHOOK
VELCRO

PRE-ENTRY ATTITUDE TIMELINE

HOOK
VELCRO

K4B

K4B

10/13/72

11/2/72

P61

06 61 L.AT _____

LO.NG _____

LIFTUP -0000T

PRO (41:30)

06 60 GM.AX _____, V400K,

γ EI _____

PRO (EMS)

16 63 RTG.0 _____ (_____)

VIO. _____ (_____)

PRO DET=60-TFE+RET.05G

RET.05G _____

P62

50 25 00041 SEP

PRO ACTIV ENTRY DAP
(ATT HOLD)

06 61 L.AT, LO.NG,
HDS UP(DN): +(-)

PRO (CMC GUIDANCE)

06 22 (X AXIS >45° OF VEL)

ENTRY CONTINUED

K5

POWER LOSS

10/13/72

CUT OUT
THIS PART**G&N****SCS**

TVC GMBL DR P&Y - 2

TVC GMBL DR P&Y - 2

MNA

Reconfigure →

SCS TVC - RT CMD

FDAI 1
 GMBL MTR P1&Y1
 ΔV THRUST A
 GDC/Except RSI
 BMAG 1
 SCS AUTO ΔV

*cb SPS P2&Y2 - open
 FDAI - 2, SOURCE - CMC
 RHC PWR DIR 2 - MNB
 BMAGS - RATE 2 (up)
 ✓ ΔV THRUST B - NORM
 AUTO RCS - MNB

MNB

Reconfigure →

SCS TVC - RT CMD

FDAI 2
 GMBL MTR P2&Y2
 ΔV THRUST B
 GDC
 BMAG 2

TVC GMBL DR P&Y - 1
 *cb SPS P1&Y1 - open
 RHC PWR DIR 1 - MNA
 BMAGS - RATE 1 (dn)
 ✓ ΔV THRUST A - NORM
 AUTO RCS - MNA

TVC SERVO 1 - AC2/MNB

TVC SERVO 1 - AC2/MNB

AC1

Reconfigure →

SCS TVC - RT CMD

FDAI 1
 GDC/Except RSI
 BMAG 1
 MIN IMP RHC 1
 RT CMD GPI 1
 SCS AUTO ΔV
 EMS ΔV Ltg

BMAGS - RATE 2 (up)
 FDAI - 2, SOURCE - CMC

*DURING CRIT BURNS:
 AFTER GMBL MTRS ON

AC2

Reconfigure →

TVC SERVO 2 - AC1/MNA

FDAI 2 GDC
 BMAG 2 MTVC
 RT CMD GPI 2

BMAGS - RATE 1 (dn)
 SCS TVC - AUTO
 ΔV CG - LM/CSM
 MTVC with TRIM TW

LOCATION
 MDC-1 covering EMS
 scroll ind

K5

BRAKING GATES

10/13/72

R(nm)	R(fps)	RETICLE ANGLE(deg)	R(ft)
1.50	45	.08	9000
1.00	30	.13	6000
.50	20	.26	3000
.25	10	.54	1500
.08	5	1.60	500
.05		2.70	300
.03		4.00	200
.02		8.50	100

4 PLY SHIM
VELCRO

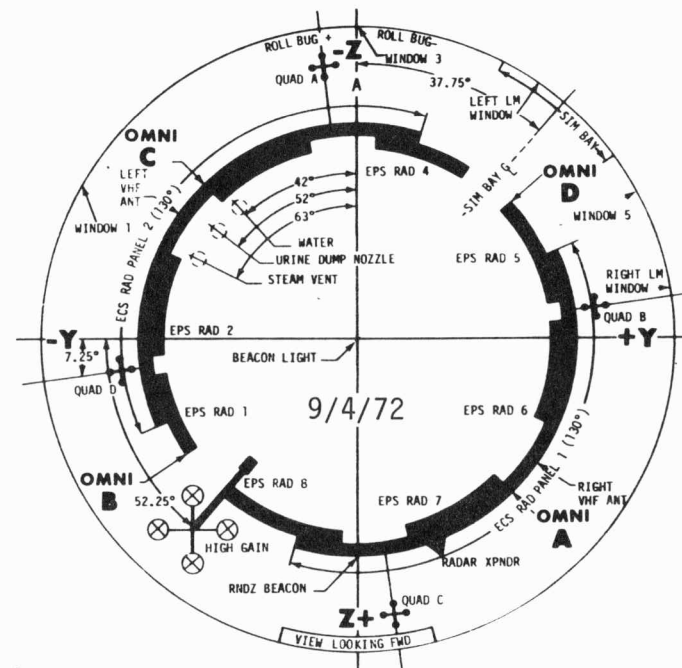
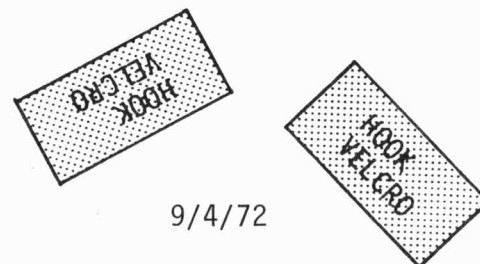
4 PLY SHIM
VELCRO

4 PLY SHIM
VELCRO

4 PLY SHIM
VELCRO

K6

K6

MNVR COMPLETION TIME

1. During auto mnvr
V4 N1 E, 3316 E
Record R1 & R2

3. V6 N34 E
(hrs, min, .01 sec)
MNVR Completion
time - 1 sec

2. V24 N25 E
Load Step 1
R1 & R2 (octal)

9/4/72

HOOK
VELCROHOOK
VELCRO

K7

K7

VEHICLE CONFIG	QUAD A/C FOR X	QUAD B/D FOR X	ERR DEADBAND	RATE SELECT
0 = No DAP	0 = Fail A/C	0 = Fail B/D	0 = $\pm 0.5^\circ$	0 = $0.05^\circ/\text{sec}$
1 = CSM	1 = Use A/C	1 = Use B/D	1 = $\pm 5.0^\circ$	1 = $0.2^\circ/\text{sec}$
2 = CSM & LM				2 = $0.5^\circ/\text{sec}$
3 = CSM & SLYB				3 = $2.0^\circ/\text{sec}$
6 = CSM & LM (Ascent Stg only)				
	N78 +90/52.25 +X = 180 -X = 000		9/4/72	
Roll Quad Select	Quad A	Quad B	Quad C	Quad D
0 = Use B/D	0 = Fail	0 = Fail	0 = Fail	0 = Fail
1 = Use A/C	1 = Use	1 = Use	1 = Use	1 = Use

LOCATION
K6 MDC-2 above
CABIN TEMP AUTO tw
K7 MDC-2 above DSKY

SIM BASIC CONFIG

9/4/72

PNL 8 AUTO RCS SELECT - OFF
 except A1,C2 (or D1,B2) A3,C4,B3,D4

PNL 3 PCM BIT RATE - HIGH
 S-BD AUX TV - SCI

PNL 181 SM/AC PWR - ON
 LOGIC POWER (2) - DPLY/RETR

PNL 230 MAP CAM ON - STBY
 MAP CAM TRACK - RETRACT
 LS OPERATE - STBY
 LS HF ANTENNA 1&2 - RETR
 IMAGE MTN - OFF
 LASER ALTM - OFF
 RCDR - HTRS
 RADAR - OFF
 MODE - HF
 DATA SYS - ON
 IR - ON
 PAN CAM SELF TEST - HTRS
 PAN CAM STEREO - STEREO
 PAN CAM MODE - STBY
 PAN CAM PWR - off (ctr)
 V/h OVRD - off (ctr)
 UV SPECT - ON
 Verify map camera retr (tb-gray)

PNL 278 MAP CAM/LASER EXP COVERS - CLOSE
 IR EXP COVER - CLOSE
 UV EXP COVER - CLOSE

PRE-SPS BURN PREP
 LASER ALTM - OFF
 ANT, CAM & COVERS -
 RETR
 LOGIC PWR (2) - OFF
 SM/AC PWR - ON
 MAP CAM ON - STBY
 IMAGE MTN - OFF
 PAN CAM PWR - BOOST

POST-SPS BURN PREP
 PAN CAM PWR - OFF
 LOGIC PWR (2) -
 DPLY/RETR

LOCATION
 ADJACENT TO PNL 230

K8

9/4/72



SIM EXP STATUS CODE

L1	SIM ATT A	MAP CAM COVER/POS B	LS HF ANT C	IR COVER D	UV COVER E
	+ +X FWD - -X FWD * NON SIM	0 CLOSED 1 OPEN/EXTD 2 OPEN/RETR	0 RETR 1 EXTD	0 CLOSED 1 OPEN	0 CLOSED 1 OPEN
L2	PAN CAM F	MAP CAM/LASER ALTM G	LS H	IR I	UV J
	0 OFF/STBY 1 PWR/STBY 2 PWR/OPERATE 3 BOOST/STBY	0 OFF/OFF 1 STBY/OFF 2 ON/ON 3 STBY/ON 4 ON/OFF 5 ON(IMC-OFF)/OFF	0 OFF 1 HF MODE 2 VHF MODE 3 RECV ONLY 4 STBY	0 OFF 1 ON	0 OFF 1 ON

USUAL CONFIGURATIONS

PRE - SPS BURN PREP (*0000)
(31000)

SLEEP (±0011)
(01011)

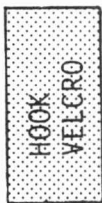
MIN POWER (0000)
(00000)

SIM BAY SECURE (0000) or (0001)
(Dumps, Thermal, Thrusters) (01011)

CM EVA

10/13/72

Don helmet, lock
 Don wrist tether, ring fwd
 Don LEVA, verify alignment
 Don comfort gloves



Don one glove, lock
 Pnl 603 EVA O2 - OFF
 Don other glove, lock
 Pnl 603 EVA O2 - ON
 Modulate OFF/ON as reqd
 to pressurize CMP
 Cuff gage 3.7-4.0 psig
 Pnl 604 SUIT PRESS ALARM - ON
 Verify tone - off
 Pnl 10 Adjust CMP Master Vol
 if required
 LMP Pnl 351 EMER CAB PRESS - OFF

EVA WARNING TONE CHECK

CMP Monitor cuff gage
 PURGE VLV - HI (CCW), verify EVA
 tone on at 3.1-3.4 psig,
 then PURGE VLV - CLOSE
 Verify tone - off
 PURGE Vlv - lock, HI
 CDR install pin
 Verify flow & cuff gage reads
 3.7-4.0 psig

CMP INTEGRITY CHECK

Pnl 603 EVA STA O2 - OFF
 Monitor cuff gage to verify PCV
 closes. Monitor press decay for
 1 min, verify <0.8 psi



(If suit press decays
 0.3 to 0.8, reverify
 helmet, wristrings,
 outer zipper and all
 connectors locked.)

Tone - on (low flow)
 Pnl 603 EVA STA O2 - ON
 Install guard
 Verify PGA press 3.7-4.0 psig
 (stable), tone - off

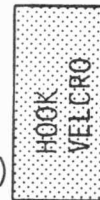
EVA STA press gage 100-500 psi
 Verify SRG TK press >750 psia

CABIN DEPRESS

GN2 vlv handle - pull
 Gage - min, leave in vent position
 Verify helmet & gloves locked
 Confirm GO for depress from Hou

NOTE: EVA warning tone may come
 on momentarily during depress

Side hatch dump vlv - open
 O2 Hi Flow lt may come on before
 cabin press reg lock up
 Close side hatch dump
 vlv at 3.25 psia
 O2 FLOW ind <0.5 LB/HR
 CDR/LMP verify suit 3.5-4.0 psia
 Side hatch dump
 vlv - open
 Dump cabin to zero



CDR/LMP verify suit
 3.5-4.0 psia (stable)
 HI O2 Flow lt - off
 CMP verify CMP cuff
 gage press 3.7-4.0 psig
 EVA STA press gage 100-500 psi,
 No tone
 Pnl 3 S-BD AUX TV - TV
 Pnl 9 MODE - INT/PTT

HATCH OPENING

Lock pin release knob - unlock
 ind released (white to yellow)
 Gear Box sel - unlatch
 ACTR handle sel - U
 Unstow ACTR handle
 Unlock Hatch
 ACTR handle SEL-L
 Stow ACTR handle
 Gear Box sel - latch
 CMP/LMP lower inner visors



Open hatch
 Verify Hex clears
 Hatch full open

LOCATION

MDC-1 covering ABORT
 and LV light panel

CM EVA CONTDEVA OPERATIONS

11/15/72

Egress

Lower outer visors

Verify TV quality

CMP adj aperture if reqd



DAC: (T8,1/250,3)

6 fps

TV: Zoom 25, Focus 4,
Aperture 22

Install/align TV/DAC pole

Jettison jett bag

Pnl 15 UTILITY PWR - ON

DAC - on, verify DAC operation

Transfer to SIM BAY and

enter foot restraints

LMP tend umb, stop at mark

CMP verify cuff gage 3.7-4.0 psig

Retrieve Lunar Sounder cassette

Remove cover

Attach hook and lock

Rotate handle, verify locked in
detent & remove cassette

Transfer to hatch

LMP attach LMP tether large hook
to cassette and lock

Remove wrist tether hook

Temp stow cassette on LMP couch

CMP to work sta

Cuff gage 3.7-4.0 psig

Retrieve Pan Camera cassette

Remove hard and soft
covers

Attach hook and lock

Pull pip pin, squeeze
handle and remove
cassette

Transfer to hatch

LMP attach LMP tether large hook
to cassette and lock

Remove wrist tether hook

Temp stow cassette on LMP couch

CMP to work sta

Cuff gage 3.7-4.0 psig

Rest - inspect SIM BAY

Retrieve Mapping Camera cassette

Remove hard and soft covers

Attach hook and lock

Squeeze handle and rotate

Remove cassette

Transfer to hatch

LMP attach CDR tether large hook
to cassette and lock

Remove wrist tether hook

Temp stow cassette

LMP move to ctr couch area

Pnl 15 UTIL PWR - OFF

Rest at hatch

CMP remove TV/DAC pole, stow
under CDR couch

Ingress

S-BD AUX TV - OFF (ctr)

CDR Pnl 9 MODE - VOX

HATCH CLOSING

Unstow Tanyard

Pull and close hatch

Verify hex clears

Close hatch

Unstow ACTR handle

Lock hatch, verify lock pin
dropped in (white to white)

Stow ACTR handle

ACTR handle sel - N

Gear Box sel - latch, ver

CABIN REPRESSRapid repress: Use EVT DECAL

Side hatch dump vlv - close

Pnl 326 REPRESS PKG - OFF

Pnl 601 REPRESS 02 - OPEN,
close @ cabin press 1 psia

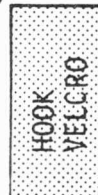
Monitor cab gross leak (30 sec)

Pnl 601 REPRESS 02 - OPEN

Monitor REPRESS 02 to zero

Cabin press ~2 psia

Pnl 601 REPRESS 02 vlv - CLOSE

CMP umb flow to
cab to 3.0 psiaGo to Cue Card
CM POST EVA

K10

10/13/72

K10

CM POST EVA

10/13/72

HOOK
VELCRO

CMP discon OPS 02 hose, hold
(direct into open volume)

OPS 02 - on

Hi 02 Flow lt - on

Monitor cabin press to

5.0 psia

OPS 02 - OFF

Hi 02 Flow lt - off

Pnl 9 MODE - INT/PTT

POST EVA

Pnl 351 EMER CAB PR - BOTH

CDR/LMP doff gloves, helmets

Pnl 380 SUIT CKT

RTN Vlv - open (pull)

Unstow Accessory bag

(CDR TSB, top pkt)

CMP PURGE VLV - HI, pull

pin & activate PURGE VLV

Stow pin in accessory bag

Pnl 603 EVA STA 02 - OFF

Pnl 604 SUIT PR ALARM - OFF

When PGA press zero psid,

CMP pop glove

DO NOT REMOVE GLOVES

Pnl 603 EVA STA 02 - ON

CMP place both gloves in
accessory bag

Remove gloves using wrist
rings only

Doff helmet with LEVA

Pnl 603 EVA STA 02 - OFF

Verify SURGE TK press >400

Pnl 326 REPRESS PKG - FILL

Go to CSM EXP/EVA C/L

page X/3-19

HOOK
VELCRO

HOOK
VELCRO

LOCATION

MDC-2 on left side
of FDAI

CM EVA MALF

9/4/72

1 TONE ON

OPS - ON

Tone on:

Pnl 603 press 65-500 (WARNING TONE MALF)

Pnl 603 press not 65-500:

PURGE VLV - HI

PCV - CLOSE, SCU - CLOSE

Pnl 603 press <65: EVA STA 02 - OFF

(Umb or CM leak or supply blkd)

Pnl 604 - OFF, Tone off

Tone off:

Cap PRV, OPS - OFF

PGA press 3.7-4.0 (PRV failed open)

PGA press not 3.7-4.0:

OPS - ON

(PCV failed open or PGA leak)

2 CUFF GAGE <3.4 & NO TONE

OPS - ON

PGA press not 3.4-4.0

(Gage failed)

PGA press 3.4-4.0:

(Tone failed)

Pnl 603 press 65-500:

Cap PRV, OPS - OFF

PGA press 3.7-4.0

(PRV failed open)

PGA press not 3.7-4.0:

OPS - ON

(PCV failed open or PGA leak)

Pnl 603 press not 65-500:

PURGE VLV - HI

PCV - CLOSE, SCU - CLOSE

EVA STA 02 - OFF

(Umb or CM leak or PGA inlet blocked)

3 CUFF GAGE >4.0

OPS - ON

PURGE VLV - HI, SCU - CLOSE

PGA press decr (PCV failed closed)

PGA press had been steady:

PURGE VLV - CLOSE

SCU - OPEN, OPS - OFF

PGA press still >4.0 (Gage failed)

PURGE VLV - CLOSE

SCU - OPEN, OPS - OFF

CM EVA MALF CONTD 9/4/72

4 LOSS OF COMM FROM EVA

Verify S-BD MODE VOICE not in RELAY

Pnl 10 VOX SENS - 9

Ck VOLUME tw's

Pnl 10 MODE - INTERCOM/PTT

Voice OK (Pnl 10 VOX failed)Voice not OK:

Pnl 10 AUDIO CONT - BACKUP

Voice OK:

Pnl 9 MODE - VOX

VOX SENS - as reqd

VOLUME tw's - as reqd

(CMP AUDIO CTR failed)

Voice not OK: (Personal comp failed)

Suit press tone may be inoperative

Hand signals:

1. Two tugs on umb:
Look at LMP
2. CMP points to ear,
Thumbs
up: can hear
dn: cannot hear
3. Monitor Pnl 603 gage

5 EVA COOLING INADEQUATE,
CUFF GAGE NORMAL

Pnl 603 press low, <100psi:

OPS - ON

SCU - CLOSE, PCV - CLOSE

PURGE VLV - HI

EVA STA 02 - OFF (System leakage)

Pnl 604 - OFF, Tone off

Pnl 603 press high, >500psi:

(Restriction in system)

Monitor PGA press often

OPS - ON

PURGE VLV - HI, as reqd

Pnl 603 press norm, 100-500psi:

(EVA crewman working too hard

or system leakage)

Monitor PGA press often

CONTINGENCY EVA**A CM PREP**

11/2/72

1 Unstow from A8:

Sect 1 EMU Maint Kit &
 (2) 02 Interconnects
 Sect 3 Tool kit
 Side A8 (3) MDC Bars

HOOK
VELCRO

- 2 C & R SUIT FLOW - OFF
- 3 C hoses stowed around
 CDR X-X foot strut,
 R hoses on LMP couch
 Intercon C & R 02 hoses

- 4 Install EVA Stabilizer Strut
- 5 TSB's installed on L Girth Ring
 and R & L LEB
- 6 Verify Tool Kit Jack Screws
 fully opened inside, snap to
 LH Girth Ring
- 7 Stow EMU Maint Kit in CMP
 TSB, top pkt
- 8 Hatch counterbalance:
 Pull Pip Pin & stow in
 R-6 or disconnect Hex with
 Tool B
- 9 Stow MDC Ingress bar
- 10 Install MDC bars & mtr covers
 (4 EVA Bag, A2)

B CABIN PREP

- 1 Depress tunnel, if docked
- 2 Stow optics & COAS
- 3 Stow cameras & bkt in TSB
- 4 Set up COMM panels
- 5 Stow helmet shield
 (PGA bag) in Jett bag
- 6 Disconnect PGA bag
- 7 Verify BUSS's covers
 installed & snapped
- 8 Stow BUSS's in PGA bag
- 9 Remove center couch
- 10 Close/lock marmon clamps
- 11 Stow under LMP couch, using
 2 straps (R5)
- 12 Remove PGA (PGA bag), place on
 LMP couch

HOOK
VELCRO

- 13 Disconnect PGA bag and stow
 under LMP couch
- 14 Stow RHC 1 in F2
- 15 L & R couch, stow pans, 0°
- 16 LH X-X Strut: (Connected or
 disconnected & tie off)
- 17 Stow in Jett Bag: A9 contents
 & partition, misc waste
- 18 Unstow B5 & B6
- 19 Prep Jett bag, secure
 draw string, tape

**CUT OUT
THIS PART****C SYSTEM PREP**

- 1 Pnl 325 CAB PRESS
 REL vlv (2) - NORM
- 2 Pnl 326 REPRESS PKG
 vlv - FILL
- 3 Pnl 326 SURGE TANK
 vlv - ON
- 4 Pnl 351 CAB REPRESS
 - OFF
- 5 Pnl 600 EMERG 02 vlv - CLOSED
- 6 Pnl 601 REPRESS 02 vlv - CLOSED
- 7 Pnl 602 REPRESS 02 865-935 psi
- 8 Pnl 2 CRYO PRESS - SRG/3
- 9 Pnl 2 SURGE TANK PRESS 865-935
- 10 Pnl 2 CAB FANS - OFF
- 11 Ck LM Status, prep for egress
- 12 Stow loose items

HOOK
VELCRO**D PLSS/COMM CK (if reqd)**

- 1 LM request:
 VHF AM A - DUPLEX
 VHF AM B - off (ctr, verify)
 VHF RANGING - OFF (verify)
 Pnl 9 VHF AM - T/R
- 2 Verify comm with 2 PLSS:
 CDR/EVCS #1, then LMP/EVCS #2
 OR 1 PLSS: EVCS #1 or #2

HOOK
VELCRO**E EXT LTS**

- 1 RUN/EVA-on(up)
- 2 RNDZ/SPOT-off

CONTINGENCY EVA CONTD

11/2/72

F FINAL CABIN PREP

- 1 Don PGA
- 2 L O2 hoses connect RR/BB
& locked
- 3 PGA FLOW DIVR vlv
(2) - as reqd
- 4 Zipper - lock/lock
- 5 Unstow helmet
- 6 Feedport cover locked

HOOK
VELCRO

- 7 Apply anti-fog (EMU Kit, CMP
TSB, top pkt) dry with tissue
- 8 Unzip fecal bags (2)
- 9 Don helmet, gloves & lock
- 10 Pnl 380 SUIT CKT RET vlv -
close (push)
- 11 Pnl 351 EMERG CAB PRESS - OFF
- 12 Ck connections: Helmet, gloves
O2, comm, feedport
- 13 Ingress CDR couch

G PRESS INTEGRITY CK

- 1 Use DECAL - EVT
- 2 Sel cont mode, mnvr to EVT att
- 3 Adj mirror to read cab press

H CABIN DEPRESS

- 1 Use DECAL - EVT

I HATCH OPENING

- 1 Use DECAL - EVT

J EVT STATION KEEP

- 1 Jett Bag, B5 & B6
- 2 AUTO RCS SEL - undock xfer
Pitch A3 & C4 - OFF
Verify 1/2° DB
- 3 AUTO RCS SEL - docked xfer
All - OFF
- 4 Mnvr as reqd for EVT

HOOK
VELCRO

K EVT

- 1 Give GO for xfer to OPS & EVT
- 2 Start DET when purge starts

L CM INGRESS

- 1 CMP protect RHC
- 2 CDR Hd 1st face MDC, sit on A9
PURGE vlv - LO
- 3 Retrieve C O2 hoses
- 4 CMP connect C elec umb to CDR
- 5 CDR AUDIO pnl as desired
- 6 Secure position, aid LMP
- 7 LMP remove xfer bag, push in CM
- 8 LMP Ft 1st face MDC, ctr couch
PURGE vlv - LO
- 9 CM back off from LM
- 10 CDR connect R elec
umb to LMP
- 11 CMP close hatch
- 12 PLSS PRIM &
AUX Feed water
vlvs - close

HOOK
VELCRO

M VAC XFER TO CM ECS

- 1 SUIT FLOW vlv - OFF
- 2 Remove interconnect

FOR OPS (O2 ON + 25 min):

- 3 Connect O2 hoses RR/BB
- 4 PURGE vlv - CLOSE
- 5 SUIT FLOW vlv - FULL
- 6 OPS O2 - OFF

FOR PLSS/OPS:

- 3 Disconnect PURGE vlv, OPS hose
- 4 Connect O2 hoses RR/BB
- 5 SUIT FLOW vlv - FULL
- 6 O2, FAN, PUMP - OFF
Mode Sel - 0

N HATCH CLOSING

- 1 Use DECAL - EVT (X/4-12)

HOOK
VELCRO

O CABIN REPRESS

- 1 Use DECAL - EVT
Go to CSM EXP/EVA
C/L X/4-7, 10, 14

EMER CAB REPRESS

9/4/72

A. CMP ON OPS Umb. flow OFF

Use EVT Cabin Repress
 decal, omit steps 3 & 4
 (gross cabin leak ck)

B. SURGE TANK Decr Rapidly

HATCH DUMP VLV - CLOSE
 OPS 02 - OFF verify
 Discon OPS Hose from PGA
 OPS 02 - ON
 CABIN PRESS ~3 psia
 P 302 SUIT FLOW - OFF ver
 Remove CMP ECS Intercon
 PURGE VLV - OPEN, HI

P 603 EVA STA 02 - OFF

P 604 SUIT PR ALARM - OFF

Discon PCV

When PGA press zero psid,

Conn ECS to PGA RR/BB

P 302 SUIT FLOW - PULL

PURGE VLV - CLOSE

P 351 CABIN REPRESS - OPEN

Maintain SRG TK 150 psia

When CABIN PRESS >4.7,

CABIN REPRESS - CLOSE

EMER CABIN PRESS - BOTH

VAC XFER TO ECSA. CMP ON OPS

P 302 SUIT FLOW - OFF ver
 Remove CMP ECS Intercon
 Discon PCV
 Discon SCU, verify CLOSED
 Conn ECS Hoses RR/BB
 PURGE VLV - CLOSE
 P 302 SUIT FLOW - FULL
 Verify flow from ECS
 OPS 02 - OFF
 Verify cuff gage 3.5-4.0
 P 603 EVA STA 02 - OFF
 P 604 SUIT PR ALARM - OFF

B. CMP ON EVA UMBILICAL

P 302 SUIT FLOW - OFF ver
 Remove CMP ECS Intercon
 OPS 02 - OFF verify
 Discon PURGE VLV
 Discon OPS 02 Hose
 Conn ECS Hoses RR/BB
 P 302 SUIT FLOW - FULL
 EVA STA 02 - OFF
 Verify flow from ECS
 P 604 SUIT PR ALARM - OFF
 PCV - CLOSE
 Verify cuff gage 3.5-4.0

K13

10/13/72

4 PLY SHIM
VELCRO

4 PLY SHIM
VELCRO

4 PLY SHIM
VELCRO

4 PLY SHIM
VELCRO

PNL 229 cb FUNCTIONS

9/4/72

PNL 229 cb	PNL 8 cb's	LOSS: RECONFIGURE
GRP 1 MNA	SCS PITCH MNA CONTR/AUTO MNA LOGIC BUS 3/4 RCS SM HTRS D PRPLNT ISOL MNA	AUTO RCS PITCH (4) - MNB Sys A Auto RCS Enable QUAD D Htrs (Att B/U) B&D He 1&2, B&D PRIM & SEC & CM RCS 1 (ISOL VLVs)
GRP 1 MNB	SCS PITCH MNB CONTR/AUTO MNB LOGIC BUS 2/3 RCS SM HTRS C PRPLNT ISOL MNB	AUTO RCS PITCH (4) - MNA Sys B Auto RCS Enable QUAD C Htrs (Att B/U) A&C He 1&2, A&C PRIM & SEC & CM RCS 2 (ISOL VLVs)
GRP 2 MNA	SCS A/C ROLL MNA SYSTEM MNA	AUTO RCS A/C ROLL (4) - MNB FDAI 1, BMAG 1 Err, Uncage, SMRD, Htrs, C&W, GDC P&R Euler: TVC SERVO PWR - AC2/MNB
GRP 2 MNB	SCS A/C ROLL MNB SYSTEM MNB	AUTO RCS A/C ROLL (4) - MNA FDAI 2, BMAG 2 SMRD, Htrs, C&W, Min Imp, GDC Y Euler, GDC Align: TVC SERVO PWR - AC 1/MNA
GRP 3 MNA	SCS YAW MNA LOGIC BUS 1/2 RCS SM HEATERS B	AUTO RCS YAW (4) - MNB QUAD B Htrs (Att B/U), SM B&D & CM 1 RCS tb's
GRP 3 MNB	SCS YAW MNB LOGIC BUS 1/4 ORDEAL MNB RCS SM HEATERS A	AUTO RCS YAW (4) - MNA ORDEAL QUAD A Htrs (Att B/U), SM A&C & CM 2 RCS tb's
GRP 4 MNA	SCS B/D ROLL MNA EMS MNA DOCK PROBE MNA SPS GAUGING MNA He VALVE MNA	AUTO RCS B/D ROLL (4) - MNB Sys A & PRIM RETRACT Bank A
GRP 4 MNB	SCS B/D ROLL MNB EMS MNB DOCK PROBE MNB SPS GAUGING MNB He VALVE MNB	AUTO RCS B/D ROLL (4) - MNA Sys B & SEC RETRACT Bank B



PNL 229 cb FUNCTIONS CONTD

9/4/72

PNL 229 cb	PNL 8 cb's	LOSS: RECONFIGURE
GRP 5 MNA	SCS DIRECT ULL MNA CONTR/DIR 1 MNA CONTR/DIR 2 MNA RCS CM HEATERS 1 RCS LOGIC MNA	1/2 Dir U11, Y D3/B4 1/2 RHC 1 DIR in A/B, all in A 1/2 RHC 2 DIR in A/B CM Eng 11-16 Htrs Ring 1 Auto RCS Auto/man Xfer, Prplnt Dump/Purge & redundant SPS Abort Ullage, CM Eng 11-16 Htrs Bank A
	SPS PILOT VLVS A	
GRP 5 MNB	SCS DIRECT ULL MNB CONTR/DIR 1 MNB CONTR/DIR 2 MNB RCS CM HEATERS 2 RCS LOGIC MNB	1/2 Dir U11, P C3/A4 1/2 RHC 1 DIR in A/B 1/2 RHC 2 DIR in A/B, all in B CM Eng 21-26 Htrs Ring 2 Auto RCS Auto/man Xfer, Prplnt Dump/Purge & redundant SPS Abort Ullage, CM Eng 21-26 Htrs Bank B
	SPS PILOT VLVS B	
BAT BUS A	SPS PITCH 1 BAT A YAW 1 BAT A EDS 1 BAT A ELS/CM SM SEP BAT A FLOAT BAG 1 BAT A SECS LOGIC A ARM A	P1 Gmb1 Mtr ON/OFF & over- current xfer to SEC Gmb1 Mtr Y1 Gmb1 Mtr ON/OFF & over- current xfer to SEC Gmb1 Mtr LV/SPS ind 1 & 3, one ABORT vote rec'd -Y Float Bag Fill/Vent Ring 1 Auto RCS Auto Xfer RCS CMD ON/OFF MN A
BAT BUS B	SPS PITCH 2 BAT B YAW 2 BAT B EDS 3 BAT B ELS/CM SM SEP BAT B FLOAT BAG 2 BAT B SECS LOGIC B ARM B	P2 Gmb1 Mtr ON/OFF Y2 Gmb1 Mtr ON/OFF LV/SPS ind 2 & 4, one ABORT vote rec'd +Y Float Bag Fill/Vent Ring 2 Auto RCS Auto Xfer RCS CMD ON/OFF MN B

K15

K15

11/22/72

LOI VGO 11/22/72

HOOK
VELCRO

SHIM - CONTINUOUS STRIP OF 4 PLY

HOOK
VELCRO

SHIM - CONTINUOUS STRIP OF 4 PLY

HOOK
VELCRO

LOI BT	VGO/THRUST	
	100%	80%
:00	2979	2979
:20	2849	2875
:40	2716	2769
1	2581	2662
1:20	2445	2554
1:40	2307	2445
2	2167	2335
2:20	2025	2223
2:40	1881	2111
3	1735	1997
3:20	1587	1881
3:40	1437	1765
4	1284	1646
4:20	1129	1527
4:40	970	1405
5	808	1281
5:20	644	1156
5:40	477	1029
6	307	901
6:20	134	771
6:35	0	672
6:40		639
7		506
7:20		370
7:40		233
8		94
8:13		0

DEC 7 LAUNCH

LOCATION

MDC-2 covering a
Portion of DSKY lights

DOI 1 LIMITS

11/2/72

ENG LIMITS	RT/ERR FOR TAKEOVER	MAN START	SHUT DOWN	EARLY C/O RESTART	RESIDUALS TRIM GUIDE
TIGHT	10/10 TERM	DELAY	BT	NO	X: to < +/- 1.0 Overburn 2.2-10: P 180, TRIM +X >10: Use SPS
If BALL VLV fails, start on suspect bank. Do not trim Y & Z residuals					

NO GO - LOSS OF:

ECS: CABIN

FIRE or SMOKE
O2 MANF (Leak)
(2) MAIN REG
PRIM COOLANT
PRIM RAD
GLYCOL (Leak)
HUMIDITY HIGH
(2) SUIT COMP
& VAC CLNR
SUIT CIR
(3) OB DUMPS
CRY: O2 or H2 TANK
EPS: (2) FUEL CELL
ENTRY BAT
MAIN BUS

EPS: BAT BUS

AC BUS
BAT RELAY BUS
(2) INV
φA AC1 or 2
G&N: AUTO ATT (2) Axis
RATE DAMP (2) Axis
DIR RCS (1) Axis
(2) BMAG P or Y
(2) FDAI
THC
(2) RHC
CMC or ISS
OSS & VHF
O-DAC
TVC SERVO LOOP
(2) DSKY

SPS: FU or OX TK (Leak)

GN2 TANK (Leak)
BALL VLV BANK
FEEDLINT T <40F
FU/OX ΔP >20
Pc <70
FLG T >480F
He TANK (Leak)
RCS: (2) SM He TK (Leak)
Leak below ISO VLV
(2) PKG T <55F
(2) P or Y THRUSTER
(2) R THRUSTER
(2) CM He TK (Leak)
& BLOWDOWN
(ARMED <1250)
CM MANF (Leak)

LOCATION

MDC-2 covering
CABIN TEMP AUTO tw

K16

9/4/72

HOOK
4 PLY SHIM
VELCRO

HOOK
VELCRO

HOOK
VELCRO

CIRC LIMITS

11/2/72

ENG LIMITS	RT/ERR FOR TAKEOVER	MAN START	SHUT DOWN	EARLY C/O RESTART	RESIDUALS TRIM GUIDE
TIGHT	10/10 TERM	START	BT+1	VGO >20	ALL <5: to <0.2 ANY >5: NO TRIM +Vgz or -Vgy: Roll L, use -Z
If BALL VLV fails, start on suspect bank.					

NO GO - LOSS OF:

ECS: CABIN

FIRE or SMOKE

PRIM COOLANT

PRIM RAD

GLYCOL (Leak)

HUMIDITY HIGH

(2) SUIT COMP

& VAC CLNR

SUIT CIR

CRY: O2 or H2 TANK

EPS: (2) FUEL CELL

ENTRY BAT

MAIN BUS

BAT BUS

AC BUS

EPS: BAT RELAY BUS

(2) INV

φA AC1 or 2

G&N: AUTO ATT (2) Axis

RATE DAMP (2) Axis

DIR RCS (1) Axis

(2) BMAG P or Y

(2) FDAI

THC

(2) RHC

CMC

ISS

OSS & VHF

O-DAC

TVC SERVO LOOP

(2) DSKY

SPS: FU or OX TK (Leak)

GN2 TANK (Leak)

BALL VLV BANK

FEEDLINT T <40F

FU/OX ΔP >20

Pc <70

FLG T >480F

He TANK (Leak)

RCS: (2) SM He TK (Leak)

Leak below ISO VLV

(2) PKG T <55F

(2) P or Y THRUSTER

(2) R THRUSTER

(2) CM He TK (Leak)

& BLOWDOWN

(ARMED <1250)

CM MANF (Leak)

LOCATION

MDC-2 covering

CABIN TEMP AUTO tw

K17

9/4/72

4 PLY SHIM
VELCRO

HOOK
VELCRO

HOOK
VELCRO